LUND UNIVERSITY
Mathematics, Faculty of Science

Number theory
MATM35
HT 2021
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Written examination
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1. Show that 2 is a primitive root modulo 19. Compute a corresponding table of indices. Use this to solve the congruences

$$x^{14} \equiv 3 \pmod{19} \qquad \text{and} \qquad 4x^{13} \equiv 5 \pmod{19}.$$

2. Given the continued fraction expansion $\frac{\sqrt{8}}{5} = [0; 1, \overline{1, 3, 3, 3, 1, 2}]$, find a convergent p_n/q_n that satisfies

$$\left| \frac{\sqrt{8}}{5} - \frac{p_n}{q_n} \right| < \frac{1}{1000}.$$

3. Prove that

$$\sum_{d|n} \frac{n\sigma(d)}{d} = \sum_{d|n} d\tau(d)$$

for all positive integers n, where τ is the divisor function and σ is the sum-of-divisors function.

4. Find an odd prime p such that all the numbers $1, 2, 3, 4, 5, 6$ are quadratic residues modulo p.

5. Prove that any prime p congruent to 1 or 3 (mod 8) can be expressed in the form

$$p = x^2 + 2y^2$$

for some $x, y \in \mathbb{Z}$.

6. Let p be an odd prime, and let r be a primitive root (mod p) with $1 \leq r \leq p-1$.
   (a) Show that either $\operatorname{ord}_{p^2}(r) = p-1$ or $\operatorname{ord}_{p^2}(r) = p(p-1)$.
   (b) If $\bar{r}$ is the unique integer such that $1 \leq \bar{r} \leq p-1$ and $r\bar{r} \equiv 1 \pmod{p}$, show that either r or $\bar{r}$ is a primitive root (mod p^2).

Give careful motivations for your solutions!